Lecture 14

Topological quantum field theory: a synthesis

Objectives. *The course has built duals (Lecture 4), F-symbols (Lecture 6), modular data (Lecture 10), algebras (Lecture 11) and centres (Lecture 12). A topological quantum field theory assigns vector spaces to spaces and linear maps to spacetimes between them. In one dimension it is determined by a dual pair. In two dimensions it is determined by a commutative Frobenius algebra. Finite groups give examples called Dijkgraaf–Witten theories. In three dimensions a spherical fusion category gives a Turaev–Viro state sum. For a unitary category its Hamiltonian form is a string net. A modular category gives the Reshetikhin–Turaev theory, related to the state sum through the centre. These constructions cover selected structures of the course. Turaev–Viro theories are non-chiral;* $\mathrm{Rep}(\mathrm{SU}(2))$ *and generic* U_q *are not finite inputs.*

By the end of this lecture you will be able to:

- Define a TQFT as a symmetric monoidal functor, and prove that its state spaces are finite-dimensional, with $Z(\Sigma \times S^1) = \dim Z(\Sigma)$.
- Define Frobenius algebras in pictures, state the classification of 2D TQFTs, and compute partition functions with the handle element.
- Compute 2D Dijkgraaf–Witten partition functions by Mednykh's formula.
- State the Turaev–Viro and Reshetikhin–Turaev constructions and their relation through the centre, with the framing anomaly, and count torus states.

14.1 Atiyah's axioms; 1D

The *bordism category* Bord_n has as objects closed oriented $(n-1)$-manifolds, and as morphisms $\Sigma \to \Sigma'$ the compact oriented n-manifolds M with $\partial M = \bar{\Sigma} \sqcup \Sigma'$, up to diffeomorphism fixing the boundary. Here $\bar{\Sigma}$ is Σ with its orientation reversed. Composition glues along the common boundary. Disjoint union $\sqcup$, with the swap of components, makes this category symmetric monoidal (Kock 1.2–1.3; Carqueville–Runkel §2). The category Cob_2 of Lecture 1 is Bord_2. Time runs upward, as everywhere in the course.

Definition 14.1 (TQFT)**.** An n-dimensional *topological quantum field theory* is a symmetric monoidal functor $Z\colon \mathrm{Bord}_n \to \mathrm{Vect}$ (Kock 1.3.32; Carqueville–Runkel Def. 2.1).

The target is *symmetric*, not braided: exchanging two spaces is the flip of vector spaces. A closed n-manifold is a bordism $\emptyset \to \emptyset$, so its image is a number, the *partition function*.

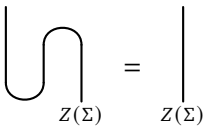
The dimension of the bordism decides what kind of value Z assigns. A surface is a spacetime in 2D and a space in 3D.

Theory	geometric input	linear-algebra output
nD	closed $(n-1)$-manifold Σ	vector space $Z(\Sigma)$
nD	bordism $M\colon \Sigma \to \Sigma'$	map $Z(M)\colon Z(\Sigma) \to Z(\Sigma')$
nD	closed n-manifold M	scalar $Z(M)\colon \mathbb{C} \to \mathbb{C}$
2D	circle S^1; closed torus T^2	space $Z(S^1)$; number $Z(T^2)$
3D	closed torus T^2; closed 3-manifold M	space $Z(T^2)$; number $Z(M)$

Thus the notation $Z(T^2)$ must always be read with the theory's dimension. Taking its dimension makes sense in the 3D row; in the 2D row it is already a scalar.

Theorem 14.2 (Atiyah's axioms, consequences · *sketched; (i)–(ii) proved*)**.** *(i) Each $Z(\Sigma)$ is finite-dimensional, with $Z(\bar\Sigma) \cong Z(\Sigma)^*$. (ii) $Z(\Sigma \times S^1) = \dim Z(\Sigma)$. (iii) A one-dimensional TQFT is determined, up to isomorphism, by the finite-dimensional space $V = Z(\mathrm{pt}_+)$, and every such V occurs (Carqueville–Runkel Thm. 3.2).*

Proof. (i) The cylinder $\Sigma \times [0,1]$, bent, is a bordism $\bar\Sigma \sqcup \Sigma \to \emptyset$ and one $\emptyset \to \Sigma \sqcup \bar\Sigma$; straightening the bend is a diffeomorphism, so their images ev and coev satisfy the snake equations of Definition 4.1. In one dimension these are pictures of the course's cups and caps:



(14.1)

Write $\mathrm{coev}(1) = \sum_{i=1}^{k} a_i \otimes b_i$, a finite sum, with $a_i \in Z(\Sigma)$ and $b_i \in Z(\bar\Sigma)$. The snake equation says $v = \sum_i a_i \,\mathrm{ev}(b_i \otimes v)$ for every $v \in Z(\Sigma)$, so $a_1, \dots, a_k$ span $Z(\Sigma)$; and ev identifies $Z(\bar\Sigma)$ with $Z(\Sigma)^*$. (ii) $\Sigma \times S^1$ is the cylinder with its ends glued, and the gluing passes one copy of Σ past the other: it is $\mathrm{ev} \circ P \circ \mathrm{coev}$, with P the flip of Vect, so $\sum_i \mathrm{ev}(b_i \otimes a_i)$, the trace of the identity, which is $\dim Z(\Sigma)$:



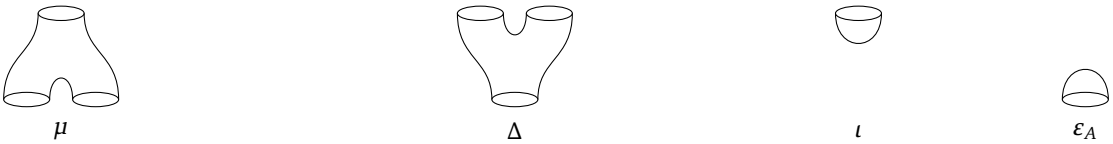
$$= \dim Z(\Sigma). \tag{14.2}$$

(iii) Every 1-manifold with boundary is a composite of the elbows and the identities, so Z is fixed by V, ev and coev; the relations between them are the snake equations, so any dual pair defines a theory. ∎

So a one-dimensional theory is Lecture 4's duality, and the circle is the loop $\dim V$ of Equation (4.7).

14.2 Frobenius algebras; 2D TQFT is commutative Frobenius

In two dimensions the objects are disjoint unions of circles, and every surface with boundary is glued from a few pieces:

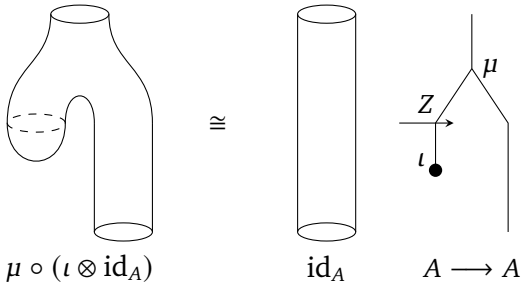


the pair of pants, read upward, merges two circles into one; turned over it splits one into two; the disc with its circle on top creates a circle from nothing; the disc with its circle below annihilates one. (Kock calls the first disc a "cap"; here the words cup and cap stay

with Lecture 4.) With $A = Z(S^1)$ they are maps $\mu\colon A\otimes A\to A$, $\Delta\colon A\to A\otimes A$, $\iota\colon \mathbb{C}\to A$ and $\varepsilon_A\colon A\to\mathbb{C}$.

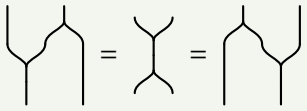
A glued surface, its graph and its map

Filling one incoming circle of the merging pants with a disc gives a cylinder. The dashed ellipse marks the glued circle, which is internal after gluing. The other incoming circle and the outgoing circle remain. The centre-line graph records the same composition after applying Z.



The surface equivalence is the unit relation in the cited presentation of Bord_2. Its image is the algebra's unit law. The graph is bookkeeping for the map; its edges stand for vector spaces, while the surface's boundary components are circles.

Definition 14.3 (Frobenius algebra)**.** A *Frobenius algebra* is $(A,\mu,\iota,\Delta,\varepsilon_A)$, an algebra (Definition 11.1) and a coalgebra on the same space (Δ coassociative with counit ε_A: the algebra axioms with the arrows reversed), satisfying the *Frobenius relation*



(14.3)

that is, $(\mathrm{id}\otimes\mu)(\Delta\otimes\mathrm{id}) = \Delta\circ\mu = (\mu\otimes\mathrm{id})(\mathrm{id}\otimes\Delta)$ (Kock Lemma 2.3.19). It is *commutative* if $\mu\circ P=\mu$, and *special* if $\mu\circ\Delta=\mathrm{id}$ (Heunen and Vicary Def. 5.5).

Equivalently, $\varepsilon_A\circ\mu$ is a nondegenerate pairing with $\varepsilon_A(xy\cdot z)=\varepsilon_A(x\cdot yz)$, and then Δ is determined (Kock 2.2.5, Props. 2.3.22 and 2.3.24). For instance $\mathbb{C}[G]$ with $\varepsilon_A(g)=\delta_{g,1}$, or the algebra of $n\times n$ matrices with the trace.

Theorem 14.4 (2D TQFTs · *cited; the bijection proved from the cited presentation*)**.** *$Z\mapsto Z(S^1)$ is a bijection, up to isomorphism, between 2D TQFTs and commutative Frobenius algebras (Abrams; Kock Thm. 3.3.2; Carqueville–Runkel Thm. 3.5, which make it an equivalence of groupoids).*

Proof. Bord_2 is generated, as a symmetric monoidal category, by the four pieces above and the cylinder, subject to relations which say exactly that they form a commutative Frobenius object: associativity and unit, coassociativity and counit, the Frobenius relation, and commutativity, which uses the swap of the two legs of the pants (Kock Prop. 1.4.13 and 1.4.24–1.4.40; Carqueville–Runkel Thm. 3.6) · *cited*. A symmetric monoidal functor therefore sends S^1 to a commutative Frobenius algebra; conversely a commutative Frobenius algebra gives values on the generators that satisfy the relations, hence a functor. ∎

Commutativity comes from the swap of the legs of the pants, which a symmetric Z sends to the flip of Vect; a Z that is only monoidal can give graded-commutative algebras (Kock

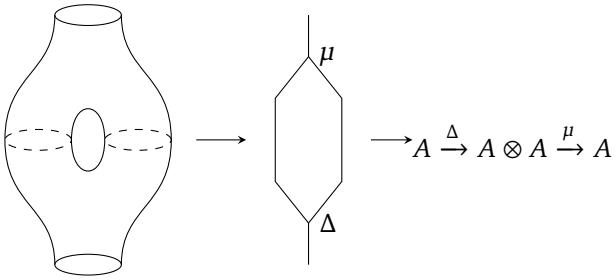
3.3.3). Every closed surface is a composite, and the genus-g surface is

$$Z(\Sigma_g) = \varepsilon_A\big(w^g\big), \qquad w = \mu \circ \Delta \circ \iota = \tag{14.4}$$



with the *handle element* w: each handle is a split followed by a merge.

The split followed by the merge has a different surface from a cylinder: it has a handle and two boundary circles. The dashed ellipses show the two circles at which the pants were glued. Its centre-line graph has a cycle.



Close its incoming circle with ι and its outgoing circle with ε_A to obtain the torus. Repeating the handle g times and closing gives the typed formula

$$Z(\Sigma_g) = \varepsilon_A \circ (\mu \circ \Delta)^g \circ \iota \colon \mathbb{C} \longrightarrow \mathbb{C}.$$

The Frobenius relation makes $\mu \circ \Delta$ multiplication by w. This explains the power w^g in (14.4). A split–merge is not generally the identity; that extra condition would be specialness.

Example 14.5 (Semisimple theories)**.** Let $A = \mathbb{C}^n$, with orthogonal idempotents $e_1, \dots, e_n$ and $\varepsilon_A(e_i) = \lambda_i \neq 0$. The pairing $\varepsilon_A(e_i e_j) = \delta_{ij}\lambda_i$ is nondegenerate, so this is a commutative Frobenius algebra, with $\Delta(e_i) = \lambda_i^{-1} e_i \otimes e_i$ and $w = \sum_i \lambda_i^{-1} e_i$. Hence

$$Z(\Sigma_g) = \sum_i \lambda_i^{1-g},$$

so $Z(S^2) = \sum_i \lambda_i$ and $Z(T^2) = n = \dim A$, as Theorem 14.2 requires. Each idempotent alone gives a theory with one-dimensional state spaces and $Z(\Sigma_g) = \lambda_i^{1-g}$, and on connected surfaces Z is their sum. Every commutative semisimple algebra over $\mathbb{C}$ is some $\mathbb{C}^n$, so this is the whole story for semisimple A, the case of the next section.

For a numerical example take $A = \mathbb{C}^2$ with $\varepsilon_A(e_1) = 1$ and $\varepsilon_A(e_2) = 2$. The unit is $e_1 + e_2$ and the handle element is $w = e_1 + \frac{1}{2}e_2$. Since the idempotents are orthogonal, the three closed surfaces give

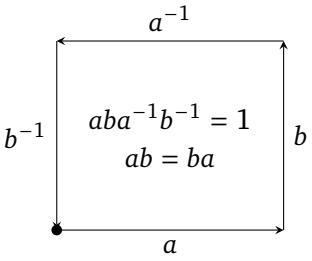
$$\begin{aligned} Z(\Sigma_0) &= \varepsilon_A(e_1 + e_2) = 1 + 2 = 3, \\ Z(\Sigma_1) &= \varepsilon_A(e_1 + \tfrac{1}{2}e_2) = 1 + 1 = 2, \\ Z(\Sigma_2) &= 1 + \frac{1}{2} = \frac{3}{2}. \end{aligned}$$

The sphere tests the counit on the unit. The torus counts the two basis idempotents. The second handle changes the value again; a closed-spacetime value need not be an integer dimension.

14.3 Dijkgraaf–Witten

A group label on a closed path records transport around that path. Following two paths multiplies their labels; reversing a path takes the inverse. Deforming a path without crossing a hole does not change its label in a flat assignment. The group $\pi_1\Sigma_g$ records closed paths from a chosen base point, with deformations identified and multiplication given by following paths. A homomorphism to G is therefore a consistent choice of such labels.

For the torus, cut along two generating paths to get a square. The boundary traversal shown below must have trivial transport when the square is filled.



The labels on opposite sides refer to the same identified path, traversed in opposite directions around the square. Thus a torus assignment is a commuting pair $(a, b) \in G^2$. More generally a genus-g surface gives tuples satisfying

$$\prod_{i=1}^{g} a_i b_i a_i^{-1} b_i^{-1} = 1.$$

For the sphere there are no such generators; the empty product is 1, so there is one assignment. This is the concrete meaning of the homomorphism count below. The surface presentation is the topological input, as in the cited Mednykh formula; no further topology is needed to enumerate these tuples.

A gauge change at the base point means conjugating every label by the same $k \in G$. The partition function divides the number of assignments by $|G|$. It is generally *not* the number of conjugacy orbits: an orbit with stabilizer $\mathrm{Stab}(a, b)$ contributes

$$\frac{|\mathrm{orbit}(a,b)|}{|G|} = \frac{1}{|\mathrm{Stab}(a,b)|}.$$

This equality is the orbit–stabilizer formula. For S_3 there are 6 elements commuting with the identity, 2 with each transposition and 3 with each 3-cycle. Consequently the torus has $6 + 3 \cdot 2 + 2 \cdot 3 = 18$ assignments and

$$Z(T^2) = \frac{18}{6} = 3.$$

The sphere instead has value $1/|G|$ in this normalization. The character formula gives a practical way to perform the same weighted count at every genus.

Theorem 14.6 (2D Dijkgraaf–Witten · *cited; the character sum sketched*)**.** *For a finite group G, the class functions on G with the convolution $(f * h)(x) = \sum_{yz=x} f(y)h(z)$ and*

$\varepsilon_A(f) = f(1)/|G|$ form a commutative Frobenius algebra, and its TQFT has

$$Z(\Sigma_g) = \frac{\#\operatorname{Hom}(\pi_1\Sigma_g, G)}{|G|} = \sum_{\chi} \left(\frac{|G|}{\dim\chi}\right)^{2g-2},$$

the sum over the irreducible characters (Mednykh; Snyder eq. (1); Freed Ex. 4.17).

Sketch. Character orthogonality (*Symmetries*, Lecture 5) gives the central idempotents as class functions:

$$e_\chi(x) = \frac{\dim\chi}{|G|}\,\chi(x^{-1}) \qquad (x \in G).$$

They satisfy $e_\chi * e_{\chi'} = \delta_{\chi\chi'} e_\chi$. Since $\chi(1) = \dim\chi$, the counit is

$$\varepsilon_A(e_\chi) = \frac{e_\chi(1)}{|G|} = \left(\frac{\dim\chi}{|G|}\right)^2.$$

In this basis $\Delta(e_\chi) = \varepsilon_A(e_\chi)^{-1} e_\chi \otimes e_\chi$, so $w = \sum_\chi \varepsilon_A(e_\chi)^{-1} e_\chi$ and $Z(\Sigma_g) = \sum_\chi \varepsilon_A(e_\chi)^{1-g}$. The first expression is Mednykh's theorem (cited; Sheet 14 proves it): it counts flat G-connections, weighted by $1/|G|$ for the gauge symmetry. ∎

For $G = S_3$, with $\dim\chi = 1, 1, 2$:

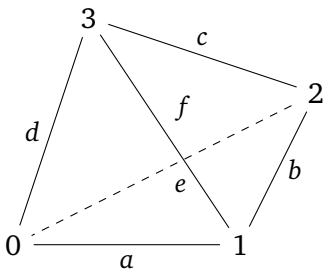
$$Z(\Sigma_2) = 6^2 + 6^2 + 3^2 = 81,$$

and $\#\operatorname{Hom}(\pi_1\Sigma_2, S_3) = 486 = 81 \cdot 6$; at genus 1 the value is 3, the number of conjugacy classes (the script counts both). Scaling ε_A by a nonzero constant multiplies $Z(\Sigma_g)$ by that constant to the power $1 - g$. Thus the counit extracting $f(1)$ instead of $f(1)/|G|$ gives a different answer.

14.4 Turaev–Viro and string nets

In three dimensions the input is a spherical fusion category C, and the F-symbols of Lecture 6 become the weights of a state sum.

The six edge labels of one tetrahedron are the six labels of a recoupling coefficient. Choose ordered vertices $0, 1, 2, 3$ and direct each edge from the smaller vertex to the larger one. The following assignment displays the correspondence with $[F_d^{abc}]_{ef}$.



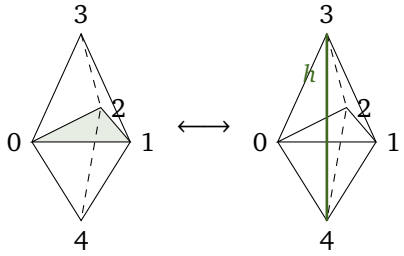
$a \otimes b \to e$
$e \otimes c \to d$
$b \otimes c \to f$
$a \otimes f \to d$

The faces 012 and 023 give the fusion tree with intermediate label e. The faces 123 and 013 give the tree with intermediate label f. Their comparison is precisely the associator with external labels a, b, c, d from Lecture 6. Reversing an edge replaces its label by the dual.

This correspondence identifies the *data* in a tetrahedral weight. It does not identify every oriented tetrahedral weight with the bare matrix entry $[F_d^{abc}]_{ef}$ in our vertex normalization. The state sum uses normalized tetrahedral evaluations, with dimension factors and duality conventions attached to the chosen face bases. The cited construction below fixes those factors consistently. The bipyramid picture then shows why the pentagon, with its intermediate label sum, is the relation needed for the 2–3 move.

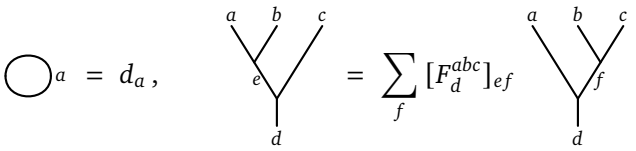
Theorem 14.7 (Turaev and Viro · *sketched*)**.** *Let C be a spherical fusion category with $\dim C \neq 0$, multiplicity-free as in Lecture 6. Colour the edges of a triangulation of a closed 3-manifold M by simple objects, weight each tetrahedron by a normalised F-symbol and each edge by its dimension, and sum over colourings with a factor $(\dim C)^{-1}$ for each vertex. (In general one also sums over bases of the fusion spaces on the faces, Turaev–Virelizier pp. 273–275.) The result $Z_{\mathrm{TV}}(M)$ does not depend on the triangulation, and extends to a 3D TQFT (Turaev–Virelizier Thms. 13.1 and 13.5; Simon (23.1)).*

Sketch. Any two triangulations are related by the Pachner moves. The 2–3 move fills the same triangular bipyramid in two ways:



The two tetrahedra share face 012. The three tetrahedra share the new edge 34. Boundary edge labels are fixed. Summing the label h of the new edge gives the intermediate-channel sum in the pentagon (Theorem 6.3), so the two sets of weights agree. The 1–4 move is absorbed by the vertex factor, by the orthogonality of the F-symbols (Simon p. 319). No braiding is used. ∎

For unitary C its Hamiltonian form is the *string net* of Levin and Wen. Put a label of C on each edge of a trivalent lattice in a surface; the ground states are the superpositions of labelled graphs invariant under local moves, the graphical calculus of Lectures 5 and 6: a loop is d_a, a bubble is removed, and the F-move of Equation (6.1) relates the two ways of splitting d into a, b, c,



(14.5)

(Levin and Wen, (4)–(7)). The Hamiltonian $H = -\sum_v Q_v - \sum_p B_p$ has commuting projectors as terms: Q_v asks that the three labels at a vertex fuse to 1, and B_p inserts the loop $\sum_s (d_s/\dim C)\, s$ inside a plaquette and fuses it into the edges with F-moves. It projects onto the subspace in which the plaquette can be filled without a puncture (Kirillov Thm. 3.9). Its coefficients are those of Lecture 10's killing loop, but the action is different: a loop is added to the lattice and resolved by F-moves. No braiding of the input C is needed. The strand-encircling projector in Lecture 10 instead uses a braiding and non-degeneracy. The ground space on a surface Σ is $Z_{\mathrm{TV}}(\Sigma)$ (Kirillov Thm. 5.1). For $C = \mathrm{Vec}_{\mathbb{Z}_2}$ it is the toric code, with four ground states on the torus (*Quantum Computing* Thm. 14.2).

14.5 RT and TV = RT ∘ $\mathcal{Z}$; the outlook

Theorem 14.8 (Reshetikhin and Turaev; Turaev and Virelizier · *cited*)**.** *(i) A modular category C with $p_+ = p_-$ gives a 3D TQFT RT_C; if $p_+ \neq p_-$ the gluing holds only up to a phase, the* framing anomaly *(Bakalov–Kirillov Thm. 4.4.3 and Rem. 4.4.5). (ii) For a spherical fusion category with $\dim C \neq 0$, $Z_{\mathrm{TV},C} \cong \mathrm{RT}_{\mathcal{Z}(C)}$ (Turaev–Virelizier Thm. 17.1, Cor. 17.5). (iii) $\dim \mathrm{RT}_C(T^2)$ is the number of simple objects of C (Bakalov–Kirillov Ex. 4.5.3).*

For a unitary C, $p_- = \overline{p_+}$, so $p_+/p_- = e^{2\pi ic/4}$ by Theorem 10.9: the anomaly vanishes exactly when $c \equiv 0 \pmod 4$, if the square root of $\dim C$ that normalises RT may be taken negative, as Bakalov and Kirillov allow. With the positive root, as unitarity asks, the gluing phase $p_+/\sqrt{\dim C} = e^{2\pi ic/8}$ is 1 exactly when $c \equiv 0 \pmod 8$. A centre has $c = 0$ (Lecture 12), so $\mathrm{RT}_{\mathcal{Z}(C)}$ is anomaly-free, as (ii) needs. Fibonacci has $c = \frac{14}{5}$: $\mathrm{RT}_{\mathrm{Fib}}$ is anomalous, and Fibonacci is not a centre, so no Turaev–Viro theory produces it alone; the closest is $Z_{\mathrm{TV,Fib}} = \mathrm{RT}_{\mathrm{Fib}\boxtimes\mathrm{Fib}^{\mathrm{rev}}}$, the doubled Fibonacci of Levin and Wen (Turaev–Virelizier Cor. 17.5 with Theorem 12.10). By (iii), the torus carries rank $\mathcal{Z}(C)$ states: 4 for $\mathrm{Vec}_{\mathbb{Z}_2}$, the toric code's degeneracy of *Quantum Computing*, and 8 for Vec_{S_3}, where 2D Dijkgraaf–Witten gave $Z(T^2) = 3$ (the script checks both).

From processes to computable invariants

In Lecture 1 the category Cob_2 described circles and surfaces as systems and processes, but no numerical value had been assigned to a surface. We can now choose a commutative Frobenius algebra, turn each disc and pair of pants into a typed linear map, and compute the value of a glued surface. The calculation for $A = \mathbb{C}^2$ above gives such a functor explicitly on its generators and evaluates its sphere, torus and genus-two surface. The cited presentation ensures that a different decomposition gives the same map.

The three-dimensional constructions use more of the course. Fusion trees describe state spaces, F-symbols change their bases, and the pentagon relates the local tetrahedral moves. The centre identifies the bulk anyons of a string net; the module-category calculation distinguishes boundaries that absorb different charges. For S_3, the value 3 of a closed torus in the 2D theory and the dimension 8 of the torus state space in the 3D theory answer different typed questions. The finite calculations are explicit; the general TV/RT and bulk–boundary constructions retain the cited or sketched status stated above. These are concrete uses of the objects, morphisms and functors with which the course began.

Optional outlook

For further study, a *fully extended* TQFT assigns data in every lower dimension, down to points. Framed theories also retain a continuously chosen frame of tangent vectors. The cobordism hypothesis classifies such theories by a higher-dimensional duality condition called *full dualizability* (Lurie, proof outline, Thm. 1.2.16). Fusion categories over $\mathbb{C}$ satisfy this condition and give framed 3D theories with a fusion category assigned to a point (Douglas, Schommer-Pries and Snyder, Thm. 1). No result here uses this classification.

Summary.

- *TQFTs:* symmetric monoidal functors $\mathrm{Bord}_n \to \mathrm{Vect}$; state spaces are finite-dimensional, $Z(\Sigma \times S^1) = \dim Z(\Sigma)$, and 1D theories are dual pairs (Definition 14.1 and Theorem 14.2).

- *2D:* commutative Frobenius algebras (Theorem 14.4); $Z(\Sigma_g) = \varepsilon_A(w^g)$ with the handle element.
- *Dijkgraaf–Witten:* $Z(\Sigma_g) = \sum_\chi (|G|/\dim\chi)^{2g-2}$; 81 for S_3 at genus 2 (Theorem 14.6).
- *3D:* Turaev–Viro from spherical fusion C, as string nets for unitary C; RT from modular C, anomalous unless $c \equiv 0 \pmod 4$, or modulo 8 with the positive root of $\dim C$; $Z_{\mathrm{TV},C} = \mathrm{RT}_{\mathcal{Z}(C)}$, and the torus counts rank $\mathcal{Z}(C)$ (Theorems 14.7 and 14.8).

References

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